

EECS 293
Software Craftsmanship
2019 Spring Semester

Quiz 1

Your name: _____

Your discussion section (time and location, or section number):

The test is 15 minutes, 10 points
Closed books and notes

1. This question is about *architectural decisions for error processing*.
 - a. (3 points) Give an example of an architectural option for error processing.
 - b. (2 points) Give a scenario where the option is appropriate and briefly explain your reasoning.

- c. (2 points) Give a scenario where the option is not appropriate and briefly explain your reasoning.

- 2. (3 points) Give an example of a *software requirement*.

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Quiz 2

Your name: _____

Your discussion leader: _____

The test is 10 minutes, 10 points
Closed books and notes

1. (3 points) Briefly explain the benefits of the comment in the following code snippet:

```
if (isValid(color)) {  
    display(color);  
}  
/*    otherwise do nothing because there is no point in  
    replacing the default background */
```

2. (3 points) Rewrite the following conditions so that the code is easier to read:

a. `while (isEligible == true)`

b. `if (rank < max_percentile && rank > min_percentile)`

c. `if (isEligible) { } else { notifyUser(); }`

3. (4 points) Briefly explain the concept of an *object factory*.

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Quiz 3

Your name: _____

Your discussion leader: _____

The test is 10 minutes, 10 points
Closed books and notes

1. (3 points) Rewrite the following code snippet avoiding any code duplication, including but not limited to the repeated invocation of `parser.lookAhead`:

```
var nextToken = parser.lookAhead(input);
while (nextToken.isFruitful()) {
    parser.advance(input);
    nextToken = parser.lookAhead(input);
}
```

2. (7 points) Calculate the cyclomatic complexity of the following code snippets. Start the count at 0. Show your work.

a.

```
if (COLOR_MIN < color && color < COLOR_MAX) {  
    y += color % 2 == 0 ? x : -x;  
} // else nothing to be done
```

b.

```
switch (cmd) {  
    case (PARSE):  
        state = Expression.parse();  
        break;  
  
    case (SCAN):  
        state = Parser.scan();  
        break;  
  
    default:  
        if ( state.equals(FAILURE) &&  
            !previousState.equals(FAILURE)) {  
            state = previousState;  
        }  
        else {  
            state = new State();  
        }  
}
```

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Quiz 4

Your name: _____

Your discussion leader: _____

The test is 10 minutes, 10 points
Closed books and notes

1. (3 points) Briefly define the concept of *information hiding*.
2. (3 points) State whether coupling should be tight or loose and briefly justify your answer.

3. (2 points) Briefly describe one advantage of using lists over arrays.

4. (2 points) Briefly outline a scenario where it is inappropriate to iterate over a list with a for-each loop.

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Quiz 5

Your name: _____

Your discussion leader: _____

The test is 10 minutes, 10 points
Closed books and notes

1. (8 points) For each of the following short descriptions, state whether the routine is a function or a procedure and give an appropriate name to the routine. In the following questions, the keyword “this” refers to the object to which the routine belongs.
 - a. Returns the last (rightmost) child of this internal node, or null if this node has no children.
 - b. Returns the tag, a private string variable associated with this node.
 - c. Returns the length of the string representation of this variable.
 - d. Convert the argument from bytes to bits.

- e. Returns whether this expression evaluates to zero.
 - f. Remove all children of this node that satisfy the given predicate. It returns true if any child was removed and false otherwise. It throws a `NullPointerException` if the specified predicate is null.
 - g. Returns whether this `SymbolSequence` appears in the given productions.
 - h. Makes the given expression a subexpression of this one. It throws a `NullPointerException` if the specified filter is null.
2. (2 points) Explain the concept of *modify parameters* to a routine, and give a simple example.

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Quiz 6

Your name: _____

Your discussion leader: _____

The test is 10 minutes, 10 points
Closed books and notes

1. (5 points) For each of the following short descriptions, state whether the routine is a function or a procedure and give an appropriate name to the routine. In the following questions, the keyword “this” refers to the object to which the routine belongs.
 - a. Change the color of this chip from black to white and sets the round index to the given argument. It returns a Boolean denoting whether this chip was modified.
 - b. Returns the coordinates of this chip’s location, a private list of integers associated with this node.
 - c. Converts coordinates from Cartesian to polar.

- d. Calculates and returns the distance of this chip from the given point on the board.
 - e. Adds this chip to the given set, and returns whether the set was modified by this operation. It throws a `NullPointerException` if the set is null.
2. (2 points) Circle the better lines of pseudo-code between the two following options, and briefly justify your answer.
- a. Set the distance of every vertex from the starting point to infinity.
 - b. `for (v: vertices) { v.setDistance(POSITIVE_INFINITY) ; }`
3. (3 points) Briefly argue in favor of creating examples of the execution of pseudo-code.

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Quiz 7

Your name: _____

Your discussion leader: _____

The test is 10 minutes, 10 points
Closed books and notes

1. (4 points) For each of the following short descriptions, state whether the routine is a function or a procedure and give an appropriate name to the routine. In the following questions, the keyword “this” refers to the object to which the routine belongs.
 - a. Compute and return the set of board locations that are adjacent to this one.
 - b. Remove all the white pebbles from this board, and returns a boolean variable denoting whether the board was changed as a result of this operation.
 - c. Returns the annotation, a private string associated with this board.

- d. Toggles all the white pebbles to black and all the black pebbles to white in this board. It returns the number of pebbles on the board.

2. (6 points) The following questions are about the use of cyclomatic complexity in the practice of software development.

- a. Give an example where a line of code of complexity zero should be moved to a separate routine, and briefly justify your answer.

- b. Briefly argue that, all else being equal, lower complexity routines are preferable to higher complexity routines, regardless of any threshold on maximum routine complexity.

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Quiz 8

Your name: _____

Your discussion leader: _____

The test is 15 minutes, 10 points
Closed books and notes

1. (2 points) After extensive unit, integration, and system testing, it was found that a software project contains 100 defects over 40 classes, and that there are two classes (out of the 40) that account for 51 defects (out of 100). Circle which one(s) of the following is a reasonable consideration or plan of action, and briefly explain your reasoning.
 - a. The project should not review defect statistics because nothing good could come out of the review.
 - b. The project is highly unusual if most defect are concentrated in so few classes.
 - c. The two most defective classes are probably designed poorly, and should be rethought and refactored.
 - d. No major refactoring is needed, but each of the two most defective classes should be split so that no single class would account for more than 10 defects.

2. (8 points) In this question, you will write test cases for the following code snippet. You are not allowed to modify the code under test. You can assume that all variables are integer.

```
while (minA <= a[i] && a[i] <= maxA) {
    i++;
}
```

- a. Write a set of test conditions on `i` and `a[i]` that ensure code and branch coverage, boundary testing, and compound boundaries. Other categories are not required. Label each condition as to whether it addresses code coverage, branch coverage, boundaries, or compound boundaries.

- b. Design at most three test cases that implement all these conditions. Label each test case with the conditions it implements. You should outline the conditions of each test and can omit the specific JUnit code.

Quiz 9

Your discussion leader: _____

1. (4 points) Briefly describe the concept of *failure atomicity* and outline one way to achieve it.

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3. (3 points) The method `public final String zipCode()` in the class `UserInterface` is supposed to return a valid zip code. However, you wish to barricade your code against erroneous user inputs, which may cause `zipCode` to return invalid zip codes or to throw exceptions. Write the code for the barricade using a status variable and specify the class or classes to which your code belongs. You should assume that you have already implemented a `final boolean isValidZipCode(String inputString)` that returns `true` if the `inputString` is a valid zip code and `false` otherwise.

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Quiz 10

Your name: _____

Your discussion leader: _____

The test is 10 minutes, 10 points
Closed books and notes

1. (4 points) Briefly explain the problem with the following code snippet and suggest a solution.

```
class Question {  
    public static final Collection<Student> roster;  
    ...  
}
```

2. (3 points) Briefly explain why an immutable class should be declared final.

3. (3 points) State whether the following statement violates the Law of Demeter and briefly justify your answer.

```
house.getDoor().getKnob().turn();
```

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Quiz 11

Your name: _____

Your discussion leader: _____

The test is 10 minutes, 10 points
Closed books and notes

1. (4 points) Briefly define the concept of *skeletal implementation* and give a short example.
2. (2 points) Give an example of a class's *self-use of an overridable method*.

3. (4 points) Consider the following two alternative definitions of SymbolSequence.

a.

```
class SymbolSequence extends ArrayList<Symbol> {  
    ParseState match(List<Token> input) { ... }  
}
```

b.

```
class SymbolSequence {  
    private List<Symbol> productions = new ArrayList<>();  
    public boolean append(Symbol symbol) {  
        return productions.add(symbol);  
    }  
    ParseState match(List<Token> input) { ... }  
}
```

Circle the better of the two approaches and briefly explain your answer.

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Quiz 12

Your name: _____

Your discussion leader: _____

The test is 10 minutes, 10 points
Closed books and notes

1. (3 points) You have found the following comment in a piece of code that you are reading. State whether the comment is appropriate and briefly justify your answer.

```
/* The following lines wrap invocations to the GoodLuck  
external library, whose use is a requirement of this  
project, but which has the tendency of throwing  
InvalidArgumentExceptions unexpectedly */
```

2. (4 points) Calculate the cyclomatic complexity of the following code snippet. Start the count at 0. Show your work.

```
MAIN_LOOP:
while (COLOR_MIN < color && color < COLOR_MAX) {
    if (isSupported(color)) {
        display(color);
    } else {
        break MAIN_LOOP;
    }
}
```

3. (3 points) For each of the following short descriptions, state whether the routine is a function or a procedure and give an appropriate name to the routine. In the following questions, the keyword “this” refers to the object to which the routine belongs.

- a. Calculates and returns the type associated with the addition of the two given types. Throws a `TypeSynthesisException` in case of error.
- b. Adds the given type conversion rule to this `TypeConversionDictionary`. If a type conversion rule already exists for the given type, no rule is added. It returns true if the operation added the given type conversion rule, and false otherwise.
- c. Returns the cardinality, a private variable of this `Operator`.